

Deals Wave — goods-management session report

2026-07-16 · branch `fix/goods-management` (unmerged) · commits `406304946` (A) · `9dd367255` (D) · `19a419159` (B) · `14fb2017e` (C) · verification record: RUNBOOK §22–§24

What the wave delivers

A complete deals vertical for the storefront, built in four phases on top of the OCS search stack: index-time **discount signals** on every product, a **Today's Deals** customer surface, admin-authored **time-boxed flash deals** with countdown/claimed-bar UX and urgency-aware ranking, a **test-driven relevance harness** that pins search quality, and **collaborative-filtering "related items"** on the product page.

Phase	Scope	Status
A	Discount signals + /deals page	Committed 2026-07-15, live-verified (125 deals, bands, sorts)
B	Flash-deal price-swap lifecycle	Committed & live-verified this session (19a419159)
C	Co-occurrence related items	Built, committed & live-verified this session (14fb2017e)
D	Relevance test harness	Committed 2026-07-15; re-run this session: 5/5 green at pinned baseline

Phase A — discount signals in OCS + Today's Deals (recap)

- Every indexed document carries `discount_pct` (counter vs retail, whole percent) and `deal_flag` (threshold `litemall.search.deal-min-pct`, default 10). CJ goods naturally index 0/0.
- Searcher scoring gained `field_value_factor(discount_pct, ln2p)` in the multiplicative stack; the search DTO carries `discountPct` so badges are data, not client math.
- Customer `/deals` page: discount bands, category chips, featured/deepest/price sorts, search-within-deals; palette v1.1 goods-list mode `deals` for the DIY home.

Phase B — flash deals via price swap (this session's close-out)

Design (ADR `docs/adr-flash-deals-price-swap.md`): the deal *is* the goods row. A 60-second scheduler (`FlashDealLifecycleTask`, kill switch `litemall.deals.lifecycle-enabled`) swaps the deal price into `litemall_goods.retail_price` when an enabled deal's window opens and restores the

captured original at swap-off (window close, disable, delete, or capped stock sold out). Checkout, cart, freight and the Phase-A discount signals therefore see the deal with **zero order-service changes** — a deliberate deviation from crmeb's separate seckill purchase path. Storage reuses the V9 `litemall_seckill` table; V38 adds only `original_retail_price + price_swapped` (the single "deal is live" source of truth).

- **Admin surface** `/srv/private/admin/deal/{list,read,create,update,delete}` — errno **650** invalid / **651** conflict (overlapping enabled window, or editing price/start while live) / **652** CJ goods refused (CJ price sync would fight the swap). `enabled` is intent, `live` is state; contract in `docs/handoff-gateway-admin-flash-deals.md`.
- **Customer:** GET `/srv/goods/deal?id=` countdown block; search DTO gains `dealActive/dealEndEpoch/dealClaimedPct` while live; "Limited-time" filter (`deal_active=1`) and "Ending soon" sort (`deal_end_epoch`) on `/deals`.
- **Ranking:** index-time `deal_urgency` (0-100 gaussian in hours-to-end) multiplied in via `ln2p`; the scheduler reindexes a live deal on ≥ 2 urgency drift. Index-time because OCS `SCRIPT_CODE` scoring takes no params.
- **Guard rails:** claimed counts recomputed each tick from paid order lines (read-only order integration); capped deals unwind early when sold out; unwind leaves an admin-edited price alone (admin wins, WARN); `counter_price` lifted to the pre-deal retail when no stronger strike-through existed.
- **SPA** (both gateways, in-worktree per Phase-A precedent): product-card countdown chip (30s tick) + claimed bar, detail-page DealBanner, admin Promotion → Flash Deals (DealList/DealForm over `adminDealApi`).

The bug this session found and fixed — OCS resolves fields against the index MAPPING.

The first full reindex ran while no deal was live, so no document carried the deal fields; the searcher then logged `ScoringCreator - Field deal_urgency ... does not exist` and **silently ignored** the `deal_active=1` filter (returned all 9,042 docs). Conditionally-emitted queryable fields are a trap: any full reindex taken during a deal-less moment disables the deal filter/sort/scoring until a post-activation searcher restart. **Fix shipped:** the three queryable fields ride every document — `deal_active:0`, `deal_urgency:0` (`ln2p(0)` = uniform baseline, ranking unchanged — proven by the harness) and `deal_end_epoch:4102444800000` (2100-01-01 sentinel, so "ending soon" ascending keeps live deals first). Result-only `deal_claimed_pct` stays live-only. The DTO maps deal fields only when `deal_active=1`, so the sentinel never reaches clients.

Live verification (worktree jar :8093 · OCS recreated from worktree compose · reindex → ocs-125, 9,042 docs)

```
create guards: CJ goods → 652 · price ≥ retail → 650 · past window → 650 · overlap → 651
activation ≤1 tick: goods 1009012 retail 59.00 → 41.30 (counter 79 kept),
seckill{price_swapped:1, original:59}
```

```
index: resultData{deal_active:1, deal_end_epoch, deal_claimed_pct:0} +
scores{deal_urgency:100, discount_pct:48}
search: deal_flag=1&deal_active=1 → total 1 with dealActive/dealEndEpoch/dealClaimedPct in
the DTO
      sort=deal_end_epoch → live deal FIRST · facet rail unchanged · zero ScoringCreator
warnings
live rules: price edit → 651 ("disable first") · stop extension → allowed · /srv/goods/deal
→ full block
unwind ≤1 tick after disable: retail/counter byte-identical restore · index back to
deal_active:0 · filter → 0 hits
regression: q=pillow total 236 + facets · suggest · home rails 6/6/3 · promotion /seckill
reads unaffected (columns only added)
```

Phase C — co-occurrence related items (built this session)

RelatedGoodsRefreshTask (nightly 02:45, `litemall.recommend.refresh-cron`; manual POST `/srv/private/admin/recommend/rebuild`) recomputes `litemall_goods_related` wholesale: **co-purchase pairs** (distinct paid-or-later orders containing both goods) weighted $\times 3$, blended with **co-view pairs** (distinct users sharing both goods in 90-day footprints — the window bounds the per-user self-join) $\times 1$. Only on-sale, non-deleted goods appear on either side; top 8 strongest-first per goods (ties id-asc, deterministic); rows whose signal vanished are pruned.

GET `/srv/goods/related` now serves the stored row first (hydration in stored order, off-sale ids drop out) and tops up / falls back to the legacy same-category query — a goods without a row degrades to exactly the old behaviour (now self-excluding). CJ branch and the response envelope are untouched, so the existing SPA related rail needed no change.

```
rebuild → {rows:15} over the dev corpus (27 paid lines / 16 footprints)
blend for goods 1010000 == manual UNION GROUP BY (12 co-view neighbours, ties → id asc, top
8 stored)
/srv/goods/related?id=1010000 → stored order top 6 · no-row id → category fallback · cj_* →
[]
```

Phase D — relevance harness re-run (regression gate for B)

The §23 judgment replay (12 queries $\times$ top-8 graded) and the four deals assertions were re-run against the live searcher after the scoring change: **5/5 green, mean nDCG@10 = 0.9286 — exactly the pinned baseline** (always-emit urgency 0 is a uniform multiplier). Runner note: mixing the platform-1.10 console/launcher with the module's 1.9.3 platform-engine dies mid-run on `TestDescriptor.getAncestors()` — put 1.10 *engine* + *commons* on the classpath too (RUNBOOK §24).

Build / ops notes

- **Silent .m2 install:** `mvn -q ... | tail; echo EXIT=$?` reports *tail*'s exit — a `litemall-db` install no-oped and the stale jar later broke packaging. Use `${PIPESTATUS[0]}` or `grep "Installing"`.
- **OCS containers mounted the MAIN checkout's ymls** (the §20 gotcha again) — recreated from this worktree's compose; they keep serving the worktree files until the merge lands the same content on master.
- **Gateways aren't in the root reactor** — build gateway-admin from its own directory. Its `webapp:test` phase is pre-existing broken (web-test-runner glob matches zero test files); the vite prod bundle builds fine.
- SPA verification = tsc (touched files clean; admin's 6 errors are node_modules noise) + prod bundles. **In-browser click-through is a post-merge step** (gateways weren't booted this session).
- `@EnableScheduling` is new to the module — `CjCatalogRefreshTask`'s documented 03:00/03:30 crons become genuinely live with this merge (they were silently inert before).

Follow-ups

- Post-merge: rebuild + restart main's :8082 goods-management, then reindex (ocs-125 already carries the always-emit fields); browser click-through of countdown/claimed bar/DealBanner/admin Deal CRUD.
- Sold-out early-unwind was proven at SQL + unwind-path level but not exercised end-to-end (no orderable checkout path this session) — worth one live pass once an orderable SKU exists in dev.
- Deals strip for the DIY home already resolves client-side via `/srv/search?deal_flag=1` (Phase A); no promotion-service work needed.